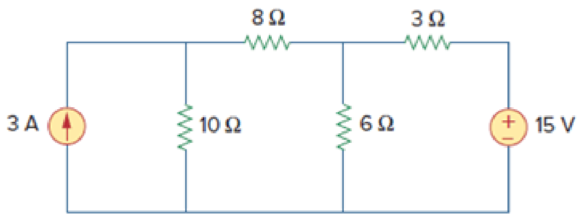
Chapter 4, Problem 23P

Problem

Referring to Fig. 4.91, use source transformation to determine the current and power absorbed by the 8-Ω resistor.

Figure. 4.91:



Step-by-step solution

Step 1 of 5

Refer to figure 4.91 given in the textbook.

Apply Source transformation technique to 3-A source and $10\ \Omega$ resistor; and apply the same to 15 V source and $3\ \Omega$ resistor as shown in figure 1.

Replace the combination of 3-A source and $10\ \Omega$ resistor with a voltage source of $(3\times10)=30\ \text{V}$ and the combination of 15 V source and $3\ \Omega$ resistor with a current source of $\left(\frac{15}{3}\right)=5\ \text{A}$ as shown in figure 1.

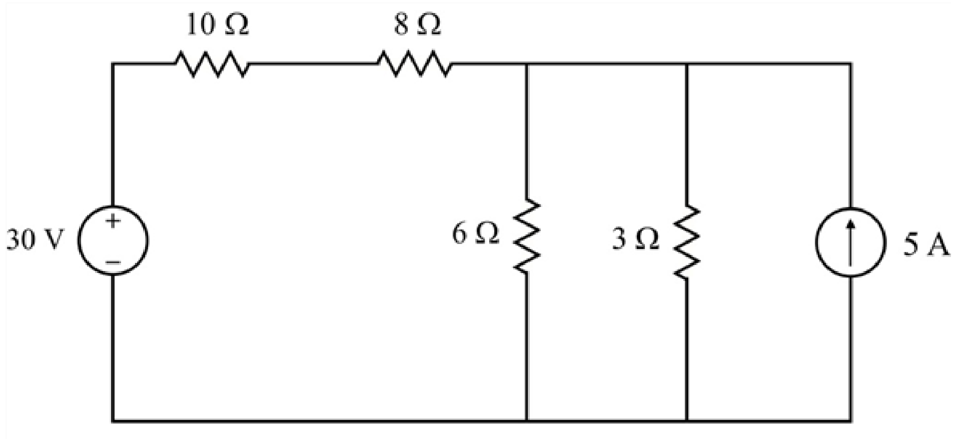


Figure 1

Step 2 of 5

Consider the circuit shown in figure 1.

The resistors $3\ \Omega$ and $6\ \Omega$ are in parallel to each other.

$$(3\ \Omega \| 6\ \Omega)=\frac{3\times6}{3+6}$$

$$(3\ \Omega \| 6\ \Omega)=\frac{18}{9}$$

$$(3\ \Omega \| 6\ \Omega)=2\ \Omega$$

Step 3 of 5

Redraw the circuit by replacing the resistances $3\ \Omega$ and $6\ \Omega$ with $2\ \Omega$ as shown in figure 2.

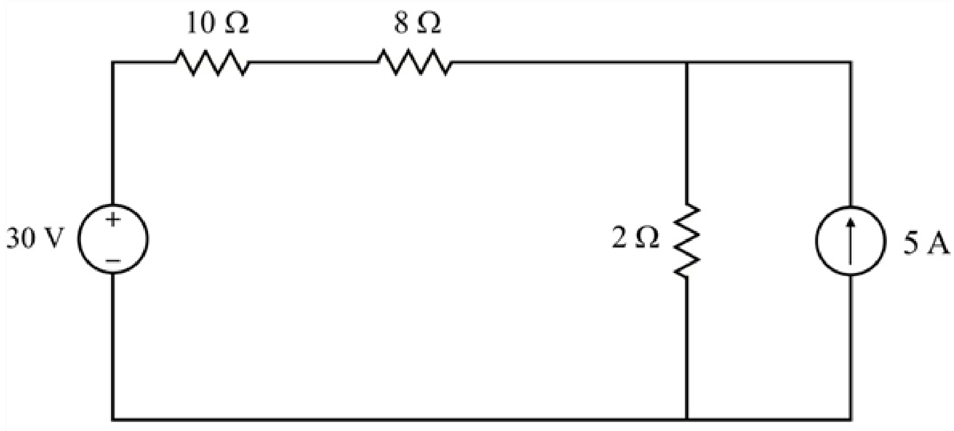


Figure 2

Step 4 of 5

Consider the circuit shown in figure 2.

Apply source transformation technique to the combination of 5 A source and $2\ \Omega$ resistor and replace the current source by a voltage source of $5 \times 2 = 10\ \text{V}$ as shown in figure 3.

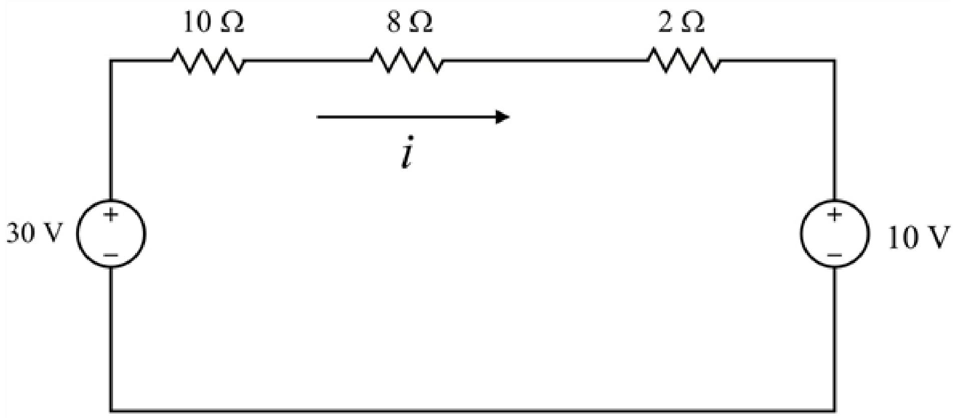


Figure 3

Step 5 of 5

Consider the circuit shown in figure 3.

Apply Kirchhoff's voltage law.

$$30 - 10i - 8i - 2i - 10 = 0$$
$$20i = 20$$
$$i = 1\ \text{A}$$

Assume that the power absorbed by the $8\ \Omega$ resistor is $P_{8\,\Omega}$.

The power absorbed by the $8\ \Omega$ resistor is,

$$P_{8\,\Omega} = i^2 \times 8$$

Substitute 1 for *i*.

$$\begin{aligned} P_{8\,\Omega} &= 1^2 \times 8 \\ &= 8\ \text{W} \end{aligned}$$

Therefore, the power absorbed by the $8\ \Omega$ resistor is $\boxed{8\ \text{W}}$.